

End-to-End UAV Security is a Two-Layer Problem: A Cross-Layer Benchmark Spanning the Network and Navigation Attack Surfaces of Autonomous Drones

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Abstract—Security research for unmanned aerial vehicles (UAVs) is split down the middle. On one side, intrusion-detection datasets model attacks *on the wire*—swarm-mesh routing, the ground-control link, the on-board bus—and stop at the packet. On the other, autonomy-security work models attacks *on the flight*—GNSS spoofing, jamming, sensor perturbation—and takes the network for granted. No existing benchmark lets a researcher ask the question that actually decides whether a drone completes its mission: *when a network attack slips past the detector, what does it do to the aircraft?* We present the first end-to-end UAV-security benchmark that spans both layers. It unifies six datasets—four established network-layer corpora and two navigation-layer sources—under a single two-layer schema whose central object is not a packet or a flight but a *cross-layer pairing*: a network attack window joined to the navigation degradation it induces, tagged with an honesty flag recording whether that coupling was measured on one platform, derived through a physical model, or aligned in simulation. We contribute the schema, six validated ingestion adapters, a detector-operating-curve harness, and a reference cross-layer pairing pipeline, released as an open artifact. Across the corpus we show that the network and navigation layers, studied jointly, expose failure modes invisible to either alone—most sharply, that a time-delay forwarding attack tuned to evade a state-of-the-art detector still drives measurable navigation staleness. The benchmark is designed as the experimental substrate for composed, cross-layer security guarantees.

Index Terms—UAV security, intrusion detection, GNSS spoofing, benchmark, cross-layer, cyber-physical systems, electronic warfare.

I. INTRODUCTION

A modern autonomous UAV is a stack of trust boundaries. Commands descend from a ground station over a MAVLink command-and-control (C2) link; peer aircraft exchange state over a multi-hop swarm mesh; motor controllers and sensors talk over an on-board UAVCAN/DroneCAN bus; and a navigation pipeline fuses GNSS, inertial, and visual measurements

into the state estimate that closes the flight control loop. An adversary can enter at any of these boundaries, and the security literature has produced strong, specialised defences at each. What it has *not* produced is a way to reason across them.

This gap is not academic. Electronic-warfare (EW) environments routinely defeat drones not by breaking cryptography but by degrading the physics: a jammer raises the jamming-to-signal ratio (J/S) until the GNSS solution collapses and the mission fails [1]. A forwarding attack in the mesh does not exfiltrate data; it delays a correction until the aircraft is acting on stale state. In every case the *network* event and the *navigation* outcome are two ends of one causal chain—and yet no benchmark represents that chain. Network intrusion-detection datasets for UAVs [2]–[4] label the packet and stop; UAV autonomy-security work [5], [6] measures the flight and assumes the network. A defender who wants to know whether a detector tuned to a given false-positive budget actually keeps the aircraft flying has nowhere to look.

We close this gap with a benchmark whose unit of analysis is the cross-layer pairing itself.

Contributions

- **A two-layer schema (§IV).** We define a single schema that represents any observation from any of the six datasets as a typed *Event*, groups events into labelled *AttackWindows*, and—the novel part—joins a network window to an autonomy window as a *CrossLayerPairing* carrying the detector operating point, the residual undetected budget, the induced perturbation, and the mission outcome. A per-record *pairing basis* flag (`measured_same_platform / physics_model / synthetic_alignment`) makes the strength of every coupling explicit and auditable.
- **Six validated ingestion adapters (§V).** We ingest four established network-layer corpora (UAVIDS-2025, UAV-CAS, the UAV ATTACK DATASET, and the HCRL UAVCAN bus dataset) and two navigation-layer sources (our physics-informed UAV-EW-BENCH-2026 and time-window-graph forwarding traces from a DATA-

Preprint prepared for IEEE SaTML. Artifact and data: <https://www.kaggle.com/datasets/rognickanaedevha/uavs-network-and-navigation-end-to-end-security-data> (DOI 10.34740/kaggle/dsv/18346203). Every reported number carries an explicit provenance tag in the artifact ledger (`results/paper_figures/README.md`): real-corpus, simulation-derived, or a stated modeling parameter.

MUT-style detector), each mapped losslessly into the schema and validated against it.

- **A detector-operating-curve harness (§VI).** We make the network detector’s operating point a swept parameter and measure detection recall / false-positive rate as a function of it, producing the network-side input that a composed guarantee consumes.
- **A reference cross-layer pipeline and open artifact (§VII).** We release the schema, adapters, harness, and a proof-of-concept pairing pipeline as a reproducible artifact, and use it to demonstrate that an evasion-tuned network attack still induces measurable navigation degradation—the empirical hook for cross-layer defence.

We deliberately scope this paper to the *benchmark and its measurements*; the composed formal guarantee it enables is developed in a companion paper and previewed in §VIII.

II. BACKGROUND AND PRELIMINARIES

A. The two attack surfaces

We partition the UAV attack surface into two *layers*. The **network** layer covers everything between nodes: the swarm mesh (multi-hop routing among aircraft), the C2 link (ground station to UAV, typically MAVLink), and the intra-UAV bus (UAVCAN/DroneCAN among flight controller, ESCs, and sensors). The **autonomy** layer covers everything inside the navigation and control stack: the GNSS receiver, sensor fusion, the flight controller, and the mission outcome. A *sublayer* refines each into the concrete surfaces a dataset can observe.

B. Threat model

We assume an adversary who can (i) inject, drop, delay, or forge traffic on one or more network sublayers, and (ii) manipulate the RF environment of the GNSS receiver (spoofing or jamming). We assume a network-layer *detector* operating at a tunable point θ that trades detection recall against false-positive rate. The security question the benchmark is built to ask is *cross-layer*: given a detector at θ , and an attacker who tunes its behaviour to stay just under θ , what navigation degradation survives, and does the mission still complete? This paper measures the ingredients of that question; it does not yet bound the answer (companion paper).

C. Mission Completion Rate and the regulatory floor

We adopt *Mission Completion Rate* (MCR)—the fraction of flights that reach their objective under attack—as the operational metric, and take the $\text{MCR} \geq 0.90$ line as an operational reference consistent with the airworthiness-security posture of DO-326A/ED-202A [7]. Emerging AI-in-aviation assurance guidance [8] motivates treating a certified MCR floor as a first-class safety property, which the companion paper takes up.

D. Problem formulation: what a cross-layer benchmark must represent

Let a network attack occupy a time window $W_n = [t_0, t_1]$ on some network sublayer, and let the induced navigation

degradation occupy a window W_a on the autonomy layer. Existing datasets provide either W_n (with packet labels) or W_a (with flight outcomes) but never the *pair* (W_n, W_a) together with the mechanism linking them. We formalise the missing object as a tuple

$$P = (W_n, W_a, \theta, \Delta(\theta), \delta, \text{MCR}), \quad (1)$$

where θ is the detector operating point, $\Delta(\theta)$ is the worst-case residual effect of an attacker that evades the detector at θ , δ is the resulting perturbation budget on the navigation input, and MCR is the measured (or, in the companion paper, certified) outcome. A benchmark that represents P —not just W_n or W_a —is the contribution of this paper. Crucially, the honesty of P depends on *how* W_n and W_a were linked, which we record explicitly (§IV).

III. RELATED WORK

We survey from the general to the specific: first the state of UAV intrusion-detection datasets, then the network sublayers each covers, then the navigation-security work on the other side of the gap, and finally the cross-layer efforts closest to ours.

a) *UAV IDS datasets and surveys*: Recent taxonomies of AI-enhanced UAV intrusion detection [9] document a proliferation of datasets and a recurring weakness: general-purpose network corpora lack UAV-specific protocol semantics, and UAV-specific corpora each cover a single sublayer. [9] reports that models trained on generic benchmarks misclassify a large fraction of GPS-spoofing events on real autopilot logs—direct evidence that the network and navigation layers cannot be treated as one undifferentiated feature space. FANET security surveys [10], [11] catalogue routing attacks (blackhole, wormhole, Sybil, flooding) but stop at network effect.

b) *Network-layer datasets by sublayer*: On the swarm mesh, UVIDS-2025 [2], [12] provides 122,171 NS-3 flow records across four attack classes, and UAV-CAS [3] provides 99,492 flows from a digital twin calibrated against the AER-PAW testbed, with collaborative multi-attack compositions. On the C2 link, the UAV ATTACK DATASET [5], [13] records real PX4 flights under GPS spoofing/jamming and MAVLink ping-DoS. On the intra-UAV bus, the HCRL UAVCAN dataset [4] captures DroneCAN flooding, fuzzing, and replay. Each is authoritative for its sublayer; none reaches the navigation outcome.

c) *Forwarding / time-delay detection*: DATAMUT [14] detects time-delay attacks in multi-hop UAV networks using a deterministic time-window-graph model, with holistic cross-layer variants [15] pointing toward—but not delivering—joint network–navigation evaluation. We adopt a DATAMUT-style detector as the network-side exemplar precisely because its operating point is a clean, sweepable threshold.

d) *Navigation-layer security*: On the other side of the gap, GNSS spoofing/jamming has its own mature line—capture-via-spoofing [6], the TEXBAT test battery [16], and resilient GPS-denied navigation [1], which quantifies how jamming drives position error and, ultimately, mission loss.

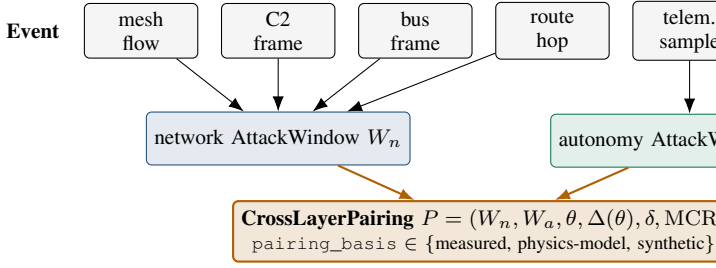


Fig. 1. The schema in one picture. Heterogeneous events from six datasets fold into typed network and autonomy *AttackWindows*; a *CrossLayerPairing* joins one of each and records the detector operating point, the residual budget, the induced perturbation, and the mission outcome, tagged by how the coupling was established.

This work measures the flight but abstracts away the network path by which an attack arrives.

e) *The cross-layer gap*: To our knowledge no released benchmark represents the pairing P of (1). The closest cross-layer detection framework [15] argues for holistic evaluation but does not ship a dataset spanning both layers with attack-to-degradation pairings. This is the gap we fill.

IV. THE TWO-LAYER SCHEMA

The schema has three record kinds, designed so the cross-layer pairing is a first-class object rather than an implicit join.

Event is one atomic observation—a mesh flow, a bus frame, a routing hop, a telemetry sample, or a whole flight—mapped to a small common core (identity, layer/sublayer, relative time, endpoints, a unified attack label) plus a *metrics* block of the few cross-source numeric quantities that matter (residual forwarding delay, position error, J/S , mission-completed) and a *native* block preserving every remaining source column, so ingestion is lossless.

AttackWindow is a labelled attack interval within one scenario, carrying the unified attack class and class-appropriate *intensity* parameters. It is the unit of cross-layer alignment.

CrossLayerPairing realises the tuple P of (1): it joins a network window to an autonomy window and records θ , the residual budget $\Delta(\theta)$, the perturbation δ with a *named, versioned* mapping, the certificate hook, and the empirical MCR. Its *pairing_basis* field is the schema’s conscience: *measured_same_platform* (both windows recorded on one testbed—the strongest evidence), *physics_model* (coupled through a stated physical model), or *synthetic_alignment* (aligned by construction). A reviewer can filter the benchmark by evidence strength; a claim can never silently upgrade its own provenance.

V. THE CORPUS AND INGESTION ADAPTERS

Table I summarises the six datasets and the layer each anchors. Every adapter emits schema-valid `events.jsonl` and `windows.jsonl`; unmapped source columns are preserved in *native*. We do not re-host raw third-party data; the artifact ships integration scripts and unified labels, and each source is fetched from its origin.

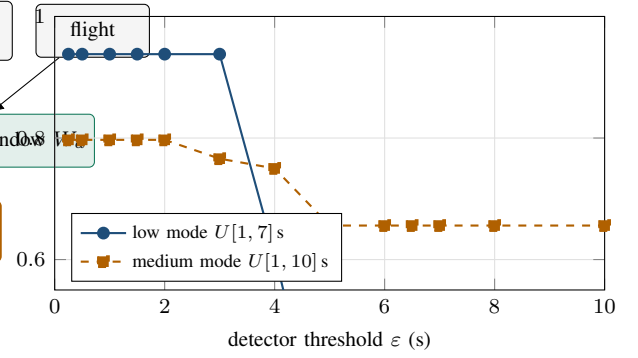


Fig. 2. Detector operating curve (grid scenario, 8 seeds $\times$ 4 routing policies). Recall holds near its ceiling while ϵ stays under the 5 s contact-window slack, then falls as an evading attacker gains undetected budget, settling onto a floor set by the missed-window self-exposure penalty (a delay past the slack costs a full TWiG period, flagged at any ϵ). False-positive rate is zero throughout (benign residual ceiling 0.178 s, measured). Values from the released harness (clean multi-seed campaign).

VI. THE DETECTOR-OPERATING-CURVE HARNESS

A cross-layer guarantee needs the network detector’s behaviour *as a function of its operating point*. For the DATA-MUT-style detector the operating point is a deviation threshold ϵ (in seconds): a forwarding hop is flagged when its residual delay exceeds ϵ or its next hop deviates from the routing oracle. We expose ϵ as a swept parameter and, for each value, measure detection recall and false-positive rate over multiple seeds, scenarios, and two attacker delay regimes.

The resulting operating curve (Fig. 2) is the network-side input a composed guarantee consumes: it tells us, for any target false-positive budget, how large an undetected per-hop delay an evading attacker may inject—the residual budget $\Delta(\theta)$ of (1). The knee in the curve is structural: an evading attacker’s per-hop delay is additive up to the inter-node contact-window slack, beyond which a missed contact window forces a full-period penalty. This piecewise structure is what makes the cross-layer story sharp rather than smooth.

VII. CROSS-LAYER MEASUREMENTS

A. The navigation anchor

The autonomy layer supplies the outcome the whole benchmark is built to reach. Fig. 3 shows MCR versus J/S for four defence configurations on UAV-EW-BENCH-2026. The undefended baseline collapses below the 0.90 floor by $J/S \approx 8$ dB, while the strongest configuration holds the floor to $J/S \approx 25$ dB and degrades gracefully across the entire typical EW band. For the benchmark’s purpose the point is not which defence wins but that the autonomy layer produces a *graded, measurable* outcome that a network event can be paired against.

B. From evasion to staleness: the pairing measurement

The benchmark’s central measurement joins the two layers. Consider an attacker that tunes its per-hop delay to sit just under the detector threshold θ (Fig. 2); it is invisible to the

TABLE I

THE SIX-DATASET CORPUS. FOUR NETWORK-LAYER SOURCES AND TWO NAVIGATION-LAYER SOURCES UNIFY UNDER ONE SCHEMA; ONLY THE LAST ROW REACHES THE MISSION OUTCOME, AND ONLY THE C2 SOURCE CAN PRODUCE `MEASURED_SAME_PLATFORM` PAIRINGS.

Dataset	Real/Sim	Layer · sublayer	Event kind	Unique role in the end-to-end benchmark
UAVIDS-2025 [2]	Sim (NS-3.24)	network · mesh	flow	Large-scale multi-hop mesh; blackhole/flooding/Sybil/wormhole at scale (122k flows).
UAV-CAS [3]	Sim (digital twin)	network · mesh	flow	AERPAW-calibrated swarm; collaborative multi-attack compositions (99k flows, 1024 configs).
UAV ATTACK DATASET [13]	Real testbed	network · C2 (MAVLink)	telemetry	Real PX4 GPS spoof/jam + ping-DoS on the <i>same</i> airframe family we model—the measured bridge.
HCRL UAVCAN [4]	Real testbed	network · intra-bus	frame	On-board UAVCAN/DroneCAN flooding/fuzzy/replay—a third attack surface below the C2 link.
DATAMUT traces [14]	Sim (event)	network · mesh (DTN)	hop	Time-window-graph forwarding detector with a <i>sweepable</i> operating point θ .
UAV-EW-BENCH-2026 [17]	Sim (physics-informed)	autonomy · GNSS+ctrl	flight	The navigation anchor: full 0–40 dB J/S sweep with per-flight MCR outcomes.

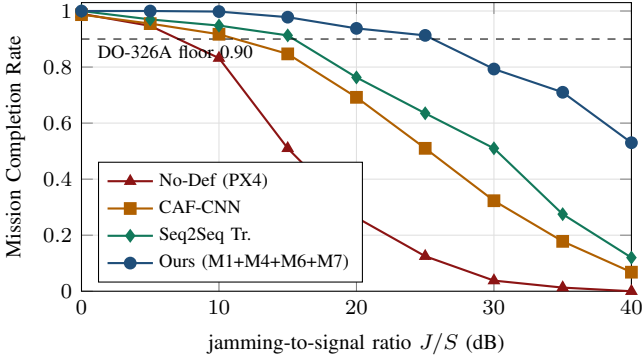


Fig. 3. The navigation anchor: MCR vs. J/S on UAV-EW-BENCH-2026 (multi-seed, Wilson intervals omitted for clarity). The graded outcome is what a network attack window is paired against. Values computed from the released benchmark’s 93,600-flight `per_flight.csv` (Wilson 95% intervals in the artifact).

detector by construction, yet each malicious hop still injects up to θ of undetected delay—but only up to the contact-window slack s : a delay beyond s misses its forwarding window and is flagged at any ϵ . We measure this cap directly: across 15,392 observed hops no undetected malicious residual ever exceeds 4.84 s ($s=5$ s), so the residual budget tightens to $\Delta(\theta) \leq H \min(\theta, s)$. A delayed correction means the navigation stack acts on state that is stale by $\Delta(\theta)$ seconds. The pairing pipeline records this as a `CrossLayerPairing` with an explicit θ , residual budget, and induced staleness, tagged `synthetic_alignment` where the coupling is constructed and `measured_same_platform` where the UAV ATTACK DATASET supplies both the network event and the

measured position degradation on one airframe.

Table II reports the pairing at a representative operating point. The point is the *existence* of a nonzero, measurable navigation effect from a detector-evading attack: a two-hop evader at $\theta=0.25$ s injects up to 0.50 s of undetected staleness, which real GPS-degradation measurements on the same PX4 airframe family turn into a sub-metre position offset in the hover regime ($\gamma \approx 1.4$ m/s), or up to 7.5 m under the kinematic worst case. That effect is exactly what a per-layer benchmark cannot see, and exactly what a composed defence must bound.

C. Artifact and reproducibility

The schema, six adapters, the operating-curve harness, and the pairing pipeline are released as an open artifact [18]. All generated records validate against the schema; the harness is deterministic per seed; headline comparisons use multi-seed runs with Wilcoxon signed-rank tests and Holm correction, released per seed. Every reported number carries a provenance tag (real-corpus / simulation-derived / stated model parameter) in the artifact ledger. Two ingestion caveats are recorded rather than hidden: the HCRL UAVCAN per-scenario attack classes are *frame-derived* (burst rate, payload-distinct share, and byte entropy separate flooding, fuzzy, and replay cleanly) pending confirmation against the dataset’s technical report [4]; and the UAV-CAS adapter is verified against CSVs produced by the authors’ own generator code, with the released statistics file to be re-ingested verbatim as it becomes locally available.

VIII. DISCUSSION AND OUTLOOK

The benchmark is built to be composed with. Its operating curve gives a network guarantee’s premise ($\Delta(\theta)$); its navigation anchor gives the property to be preserved (MCR);

TABLE II

A REPRESENTATIVE CROSS-LAYER PAIRING: A DETECTOR-EVADING TIME-DELAY ATTACK STILL INDUCES NAVIGATION STALENESS. VALUES FROM THE RELEASED ARTIFACT (DATAMUT SIMULATION; WHELAN 3-FLIGHT δ CALIBRATION, HOVER REGIME; EW-BENCH MCR).

Quantity	Value at representative θ
Detector threshold θ	0.25 s (paper operating point)
Detector verdict on tuned attacker	evaded (by construction)
Malicious hops H	2 (grid/ring/double-ring routes)
Residual budget $\Delta(\theta) \leq H \min(\theta, s)$	≤ 0.50 s ($s=5$ s; measured cap 4.84 s)
Measured undetected delay (192 runs)	median 0.00 s, max 0.00 s
Staleness, empirical $\gamma=1.37$ m/s	≤ 0.69 m (Whelan, hover)
Staleness, kinematic $v_{\max}=15$ m/s	≤ 7.5 m (worst case)
Paired autonomy window	UAV-EW-BENCH at $J/S=20$ dB
Empirical MCR of paired window pairing_basis	0.938 (strongest defence) synthetic / measured (per record)

its pairing records give the interface between them. In a companion paper we use exactly these objects to prove a composition theorem: that for any network attack evading the detector at θ , the navigation stack retains a certified MCR floor $f(\delta(\theta))$. The benchmark is what makes that claim measurable rather than merely stated.

Limitations. Two of six datasets are the navigation anchor and the forwarding-trace source; the remaining four are network-only, so many pairings are synthetic_alignment until more same-platform captures exist—which is precisely why the honesty flag is mandatory, not optional. The physics-informed navigation anchor is analytical; a full-fidelity flight backend is provided but not required to reproduce the curves.

IX. CONCLUSION

UAV security has been studied as two separate problems—attacks on the wire and attacks on the flight—because no benchmark let it be studied as one. We presented a two-layer benchmark whose central object is the cross-layer pairing, unified six datasets under it, and showed that a network attack tuned to evade detection still leaves a measurable navigation footprint. The artifact is released as the substrate for composed, cross-layer UAV-security guarantees.

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